CORPORATE FINANCE PROJECT REPORT

Smart Parking System

Near Square Hospital, Dhaka

Feasibility, Financing & Capital Budgeting Analysis

East West University | Principles of Finance (FIN101) | Section 10 | Group 03 | Summer 2026

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Dhaka parking problem: demand is concentrated where footfall is high

Part 1



What creates the business opportunity?

- Rapid urbanization and private vehicle growth increase parking demand.
- Hospitals, universities and commercial areas create recurring, high-footfall demand.
- Informal roadside parking can be insecure, unregulated and congestion-producing.
- Customers value convenience, availability and faster entry/exit, not just a parking slot.

Business implication: a reliable off-street solution can monetize convenience.

Site context: Square Hospital area

Part 1



Project location logic

Hospital proximity	200–300 m walking distance
Project capacity	100 cars + 60 motorcycles
Demand windows	8:00–11:00 AM and 3:00–7:00 PM
Core users	Patients, attendants, staff, visitors
Facility concept	Multi-level, controlled entry/exit

Demand profile: multiple customer segments, concentrated peaks

Part 1

Patients & attendants

High urgency

Need certainty and short search time

Doctors & nurses

Recurring

Potential subscription demand

Employees & visitors

Recurring / variable

Security and convenience matter

Other nearby users

Opportunistic

Incremental occupancy outside peaks

Peak demand windows

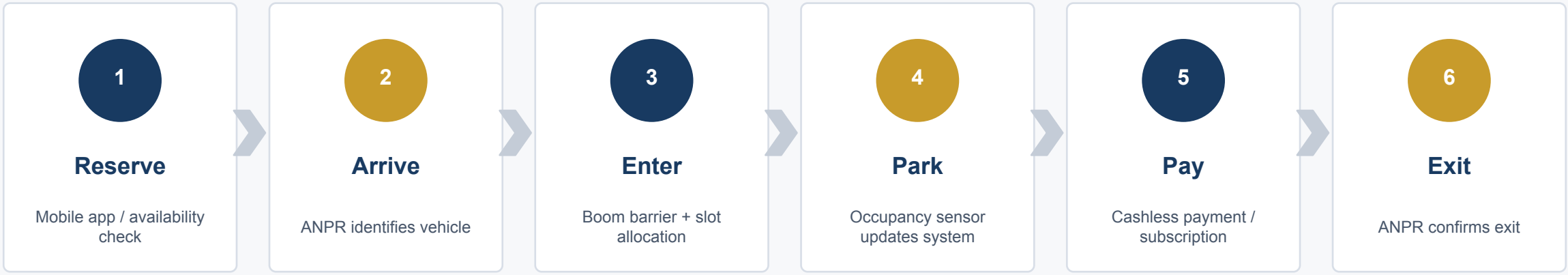
8:00–11:00 AM

3:00–7:00 PM

The operating model therefore benefits from a base of recurring demand across the day, with stronger occupancy during institutional peak periods.

Proposed solution: a technology-enabled multi-level parking facility

Part 1



Core technology layer

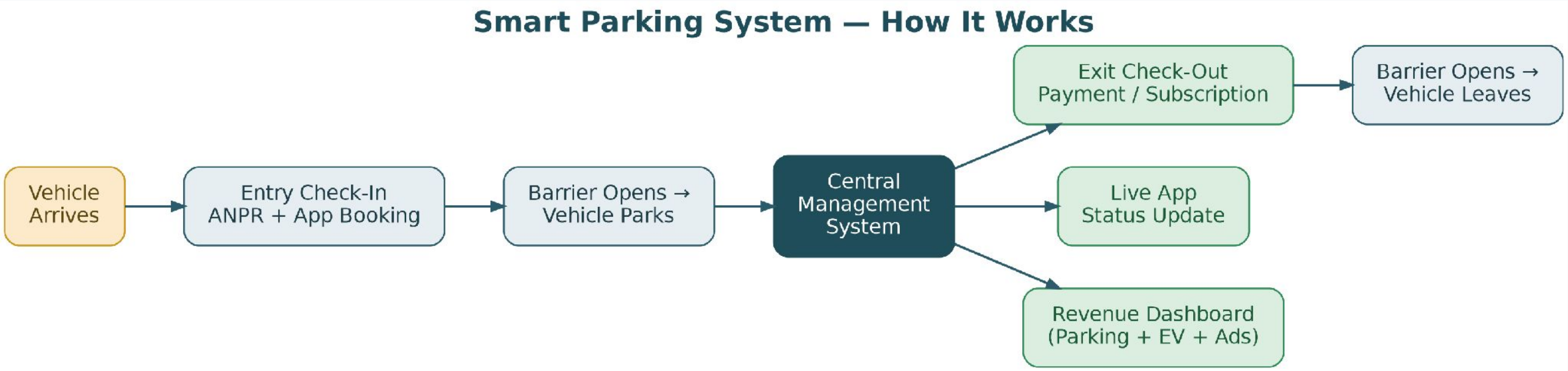
- IoT occupancy sensors
- ANPR + automated barriers
- Mobile reservation + cashless payment
- CCTV and central management dashboard

Why it matters financially

- Faster turnover can support higher effective utilization.
- Subscriptions can create recurring revenue.
- Automation can reduce friction and manual processing.

Operating workflow: from arrival to revenue capture

Part 1



Service design and capacity

Part 1

Car bays
100
Four-wheelers

Motorbike bays
60
Two-wheelers

Operating days
350
Per year assumption

Ancillary revenue
+10%
EV charging + advertising

Design principles

- Off-street, organized parking
- Controlled entry and exit
- Real-time availability and reservation
- Broad vehicle mix
- Scalable operating model

Revenue channels

Parking fees	**Primary**
Monthly subscriptions	**Recurring**
EV charging / bay rental	**Ancillary**
On-site advertising	**Ancillary**

Business objectives translated into value drivers

Part 1

Objective	Business driver	Financial implication
Safe, organized parking	Higher trust / lower friction	Supports paid occupancy
Reduce congestion	Shorter search time	Higher service attractiveness
Improve convenience	Reservation + reliable access	Supports repeat users
Technology-based management	Automation + monitoring	Scalability / operating control
Multiple revenue streams	Parking + subscriptions + ancillary	Lower single-stream dependence
Commercial viability	NPV, IRR, payback, PI	Investment decision

Market gap and competitive positioning

Part 2

Dimension	Informal roadside	Basic paid lot	Proposed smart parking
Availability visibility	Low	Low / manual	Real-time
Reservation	No	Usually no	Yes
ANPR / automated entry	No	Limited	Yes
Security / monitoring	Variable	Basic	CCTV + controlled access
Cashless payment	Variable	Possible	Core feature
Subscription model	No	Limited	Yes

Positioning: “organized + tech-enabled + convenient”

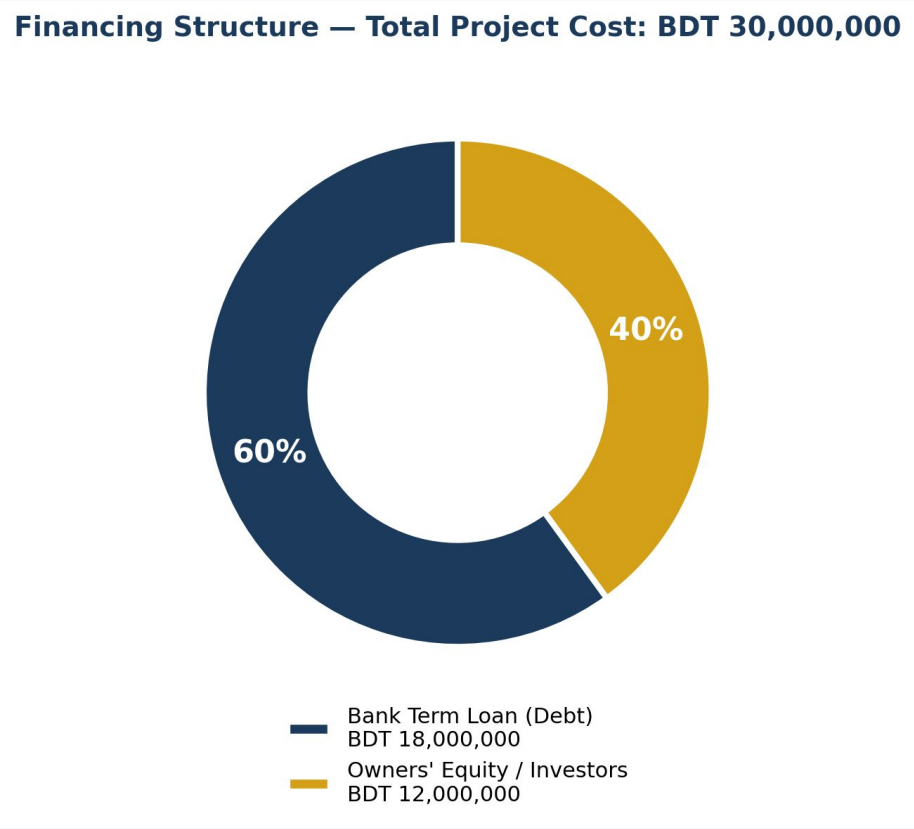
Industry and economic environment

Part 2

Vehicle & footfall growth	Parking demand remains supported by urban mobility and institutional traffic.	**Demand**
Inflation	Elevated inflation raises wages, utilities and maintenance costs.	**Cost**
Urban migration	Expansion of hospitals, universities and commercial activity supports footfall.	**Demand**
Fuel / traffic policy	Restrictions on roadside parking can shift users toward formal facilities.	**Demand**
Interest rates	Tighter monetary conditions raise project financing costs.	**Financing**

Financing plan: a blended 60/40 capital structure

Part 3



Total project financing

BDT 30,000,000

60% debt | 40% equity

- Debt finances BDT 18.0M of the project.
- Equity finances BDT 12.0M and absorbs residual business risk.
- The mix is consistent with a project expected to generate predictable cash flows.

Debt financing

Part 3

Loan amount

BDT 18.0M

60% of total investment

Loan term

5 years

Secured term loan

Interest rate

14.0%

Pre-tax

Tax benefit

10.15%

After-tax cost

Financing logic

- Secured against equipment and the lease / mortgage assignment.
- Interest is tax-deductible, reducing the effective cost of debt.
- Amortization over five years aligns debt service with the project’s operating cash generation.

Equity financing

Part 3

Equity capital

BDT 12.0M

40% of total investment

Investors

Owners + private capital

Illustratively 3–4 investors

Fixed obligation

None

Residual return profile

What equity is expected to finance

- The portion of capital not funded by debt.
- Residual cash flows after operating costs and debt service.
- Upside from project value if the facility scales or expands.

Finance principle

Equity is more expensive than debt in this model because equity investors bear more residual risk and require a higher return.

Cost of debt

Part 4

Pre-tax rate	Tax rate	After-tax cost
14.00%	27.50%	10.15%

Formula $R_d^* = \text{Pre-tax cost of debt} \times (1 - \text{Tax rate}) = 14.0\% \times (1 - 0.275) = 10.15\%$

This 10.15% figure enters the WACC calculation.

Cost of equity: CAPM

Part 4

CAPM inputs

Risk-free rate (R_f) **11.0%**
Long-term government bond proxy

Beta (β) **1.20**
Estimated project systematic risk

Market return (R_m) **16.0%**
Broad DSE market proxy

CAPM calculation

$$R_c = R_f + \beta \times [\, E(R_m) - R_f \,]$$

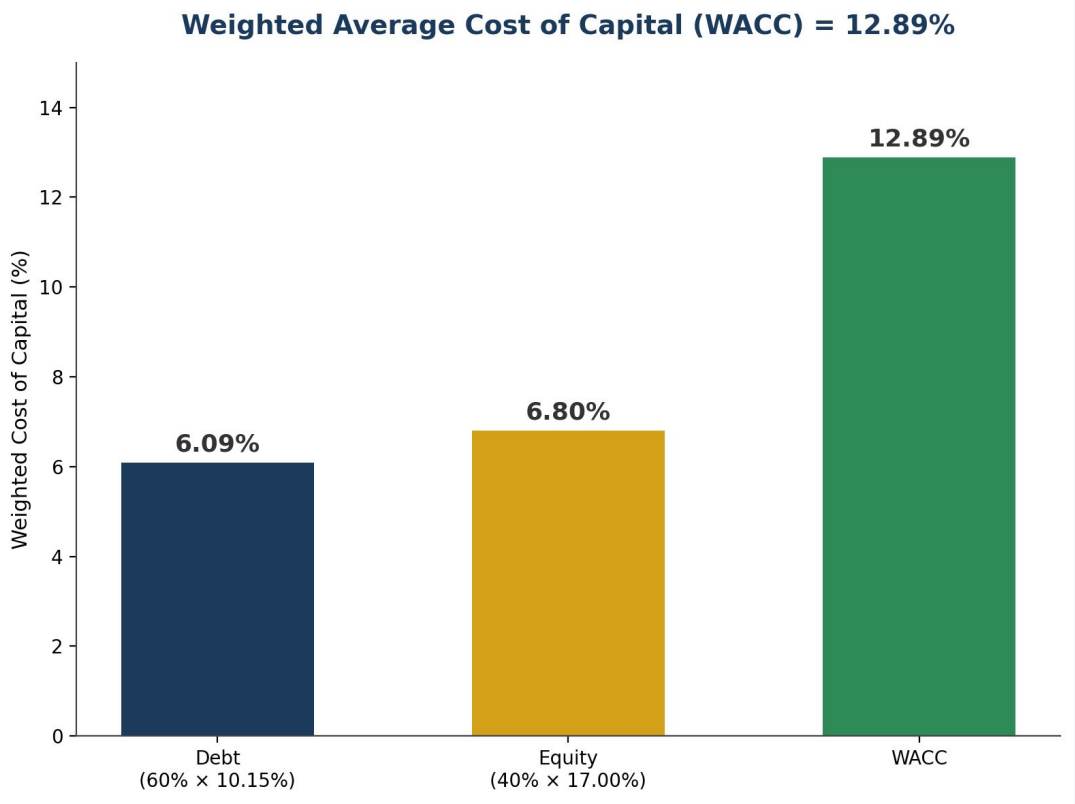
$$= 11.0\% + 1.20 \times (16.0\% - 11.0\%)$$

$$= 17.00\%$$

Required return on equity

WACC: the project discount rate is 12.89%

Part 4



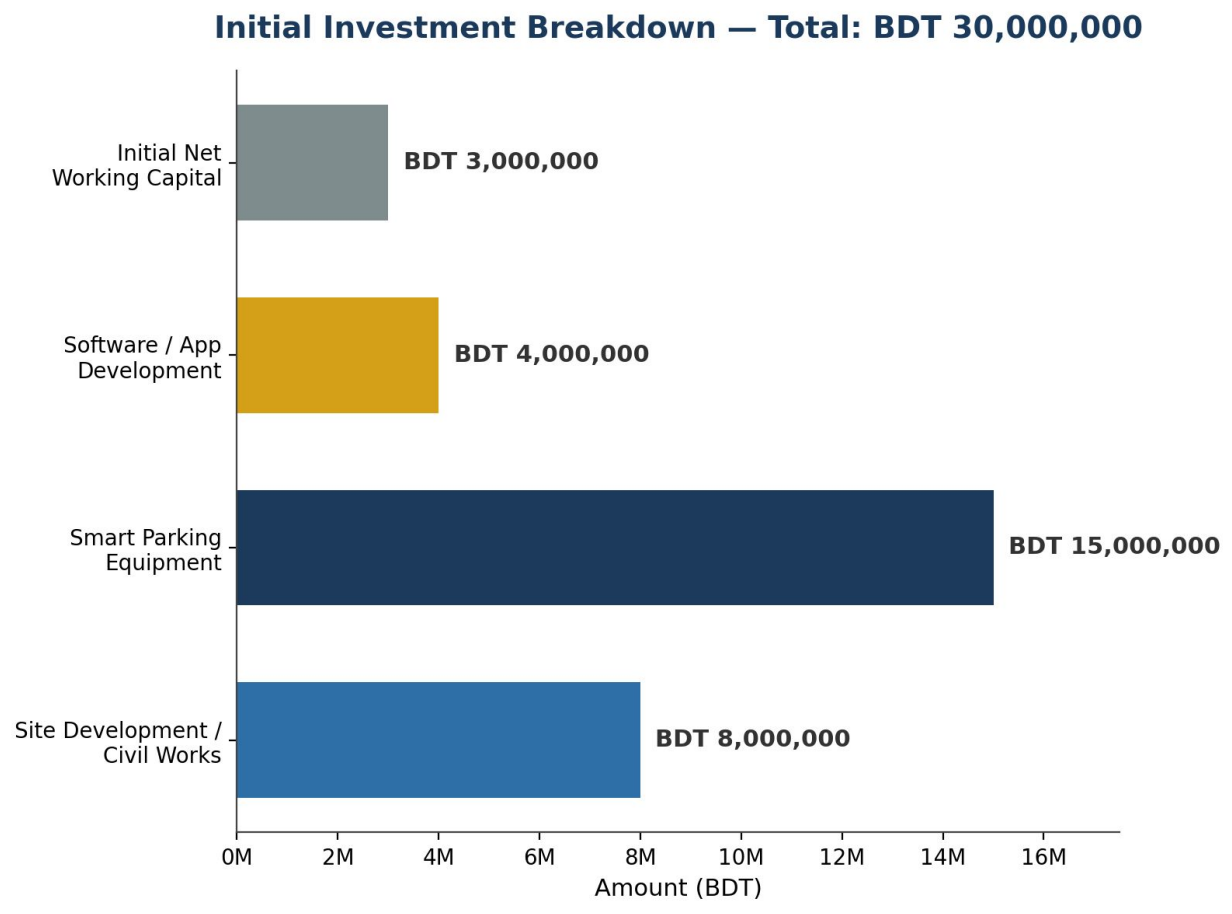
Capital source	Weight	Cost	Weighted cost
Debt	60%	10.15%	**6.09%**
Equity	40%	17.00%	**6.80%**
WACC	100%	—	**12.89%**

$$WACC = W_d R_d^* + W_c R_c$$

$$WACC = (0.60 \times 10.15\%) + (0.40 \times 17.00\%) = 12.89\%$$

Initial investment: BDT 30 million

Part 5



Use of funds	BDT
Site development / civil works	8,000,000
Smart parking equipment	15,000,000
Software / app development	4,000,000
Initial net working capital	3,000,000
Total	30,000,000

Capex concentration

Smart parking equipment = 50% of total investment

Revenue assumptions

Part 5

Car parking	100 bays × BDT 400 × 350 days	BDT 14.000M
Motorbike parking	60 bays × BDT 150 × 350 days	**BDT 3.150M**
Ancillary revenue	10% of core parking revenue	**BDT 1.715M**
Year 1 total		**BDT 18.865M**

Growth assumption **Revenue grows 8% annually, reflecting occupancy ramp-up, vehicle growth and subscription expansion.**

Core parking revenue = BDT 17.15M; +10% ancillary = BDT 18.865M

Operating cost assumptions

Part 5

Year 1 Opex

BDT 5.328M

Operating cost base

Annual opex growth

6%

Inflation-linked

Year 1 EBITDA

BDT 13.537M

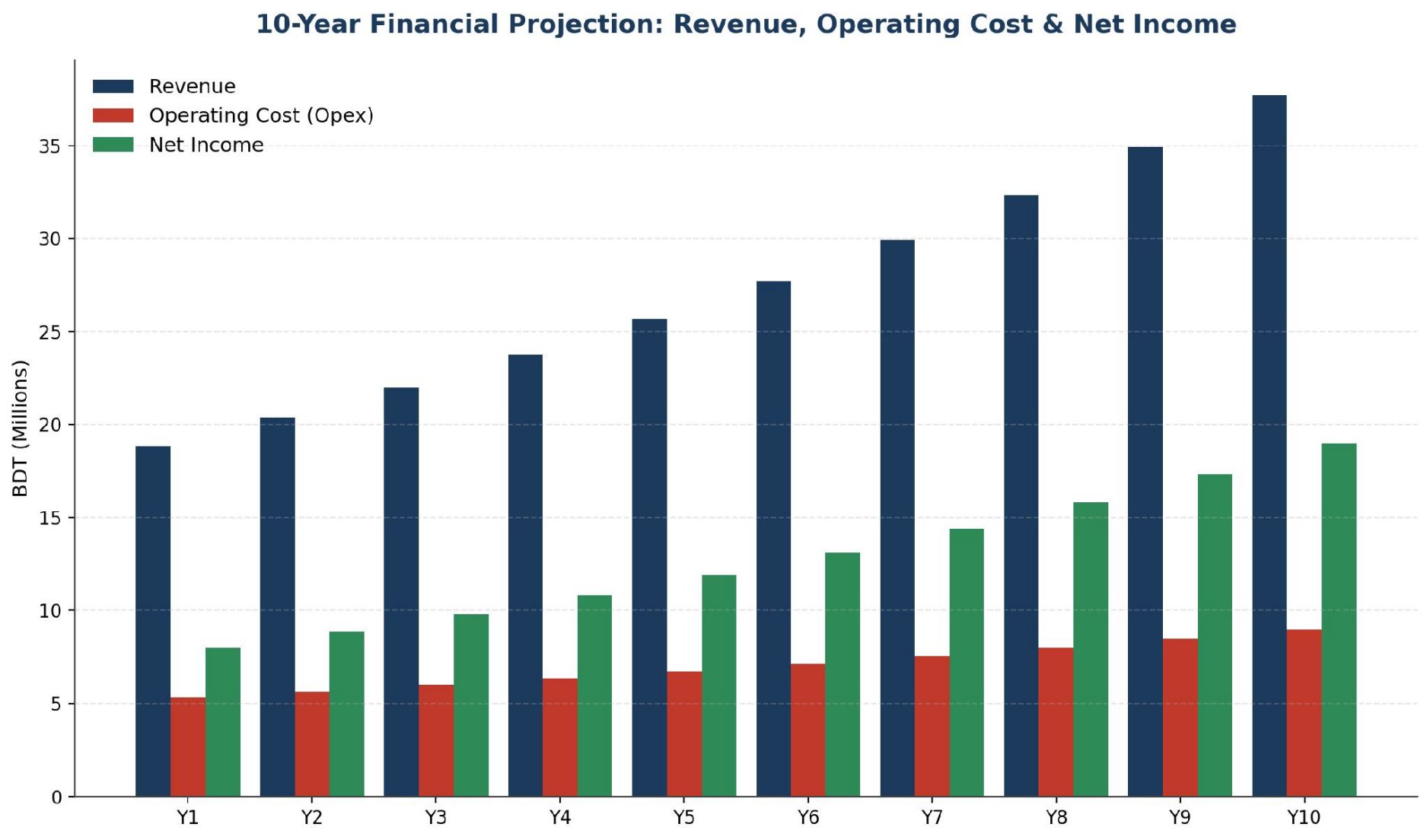
Revenue less Opex

Key cost categories

Staff costs	8 employees
Utilities	Electricity + site utilities
Maintenance	Parking equipment + premises
Software / IoT	Platform upkeep
Insurance / rent	Site and risk costs
Marketing	Customer acquisition / awareness

10-year financial trajectory

Part 5



Year 1 cash-flow mechanics

Part 5

Revenue	BDT 18.865M
− Opex	BDT 5.328M
= EBITDA	**BDT 13.537M**
− Depreciation	BDT 2.500M
= EBIT	**BDT 11.037M**
− Tax	BDT 3.035M
= Net income	**BDT 8.002M**
+ Depreciation	BDT 2.500M
= Operating CF	**BDT 10.502M**

Why depreciation is added back

- Depreciation reduces EBIT and tax-bearing profit.
- It is a non-cash accounting expense.
- Therefore, it is added back when deriving operating cash flow.

Year 1 operating cash flow

BDT 10.502M

Projected cash flow: Years 1–5

Part 5

Year	Revenue	Opex	EBITDA	Operating CF
Y1	18.865M	5.328M	13.537M	**10.502M**
Y2	20.374M	5.648M	14.727M	**11.364M**
Y3	22.004M	5.987M	16.018M	**12.300M**
Y4	23.764M	6.346M	17.419M	**13.316M**
Y5	25.666M	6.726M	18.939M	**14.418M**

Operating CF increases from BDT 10.50M in Y1 to BDT 14.42M in Y5.

Projected cash flow: Years 6–10

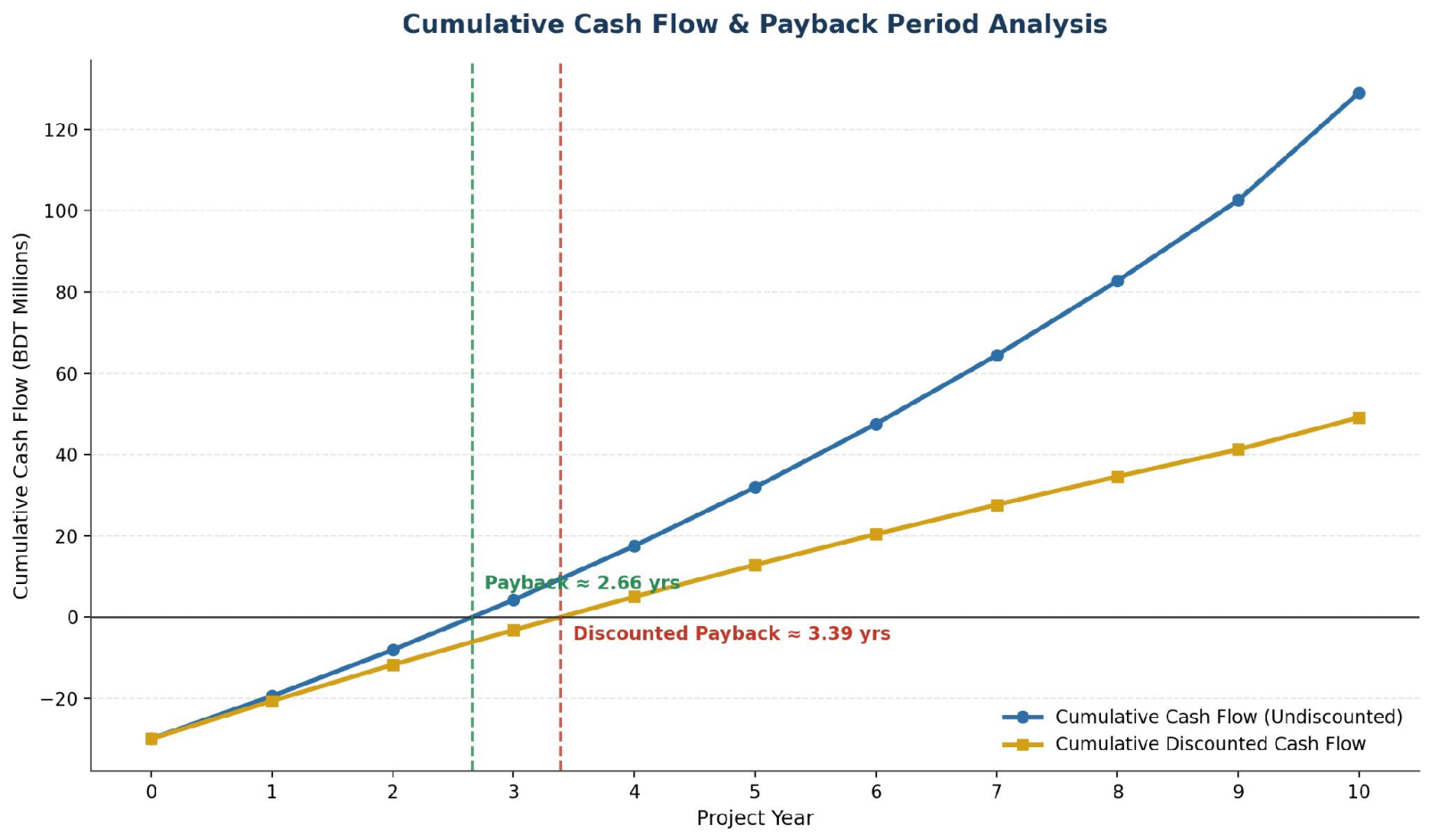
Part 5

Year	Revenue	Opex	EBITDA	Operating CF
Y6	27.719M	7.130M	20.589M	**15.614M**
Y7	29.936M	7.558M	22.379M	**16.912M**
Y8	32.331M	8.011M	24.320M	**18.319M**
Y9	34.918M	8.492M	26.426M	**19.846M**
Y10	37.711M	9.002M	28.710M	**26.502M**

Y10 operating CF includes BDT 2.0M salvage value + BDT 3.0M working-capital release.

Cumulative cash flow and payback

Part 6



Payback Period

2.66 yrs

Target < 5 years

Discounted Payback

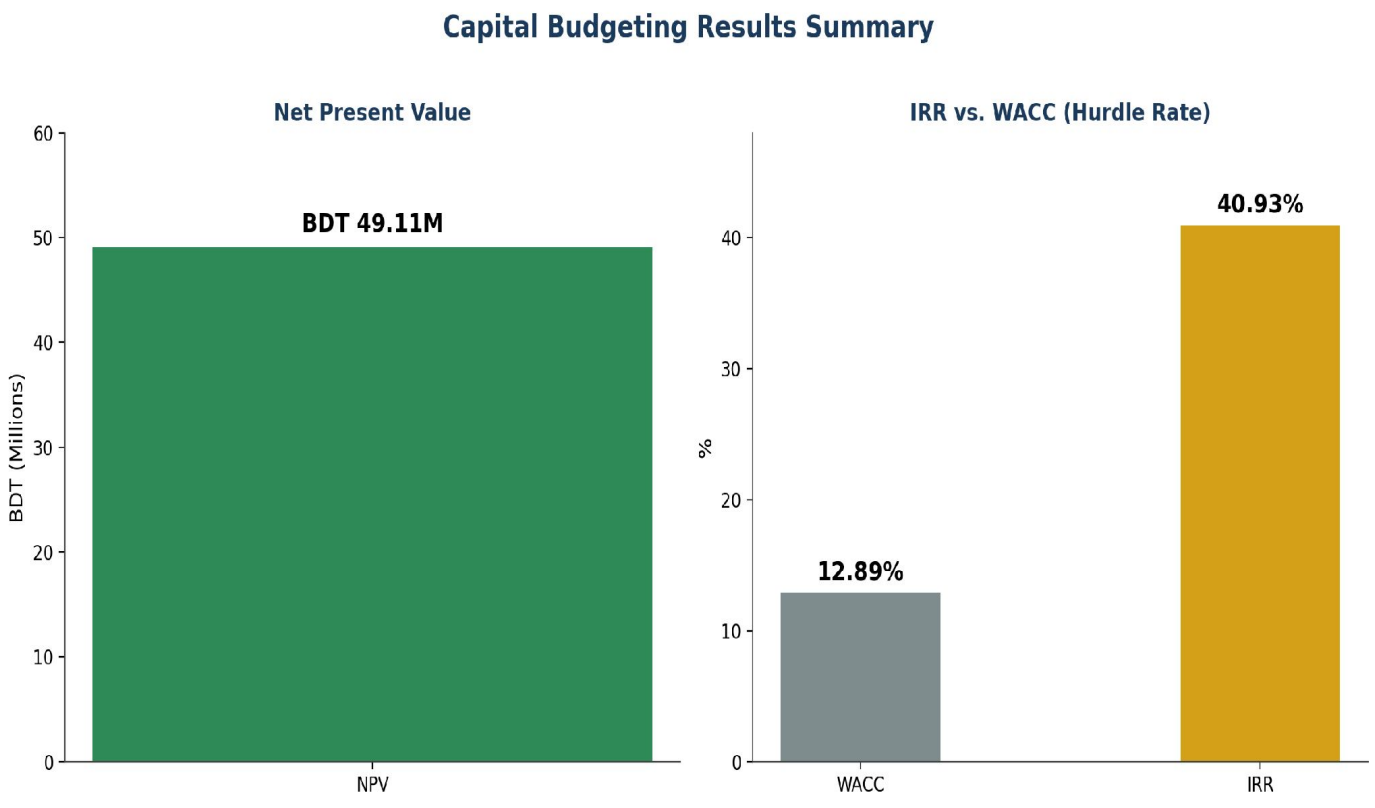
3.39 yrs

Within 10-year life

Decision: Passes both recovery tests.

Capital budgeting dashboard

Part 6



NPV	**BDT 49.107M** > 0 = Accept
IRR	**40.93%** > WACC 12.89%
Payback	**2.66 yrs** < 5 yrs target
Discounted payback	**3.39 yrs** < 10-year life
PI	**2.64** > 1.0 = Accept

NPV and IRR: the strongest value-creation signals

Part 6

Net Present Value

BDT 49.11M

Decision rule: NPV > 0

Interpretation: the project is expected to add value after covering the required return on capital.

Internal Rate of Return

40.93%

vs. WACC 12.89%

The 28.04 percentage-point spread provides a substantial buffer against moderate estimation error.

Profitability Index: value created per BDT invested

Part 6

2.64

Profitability Index

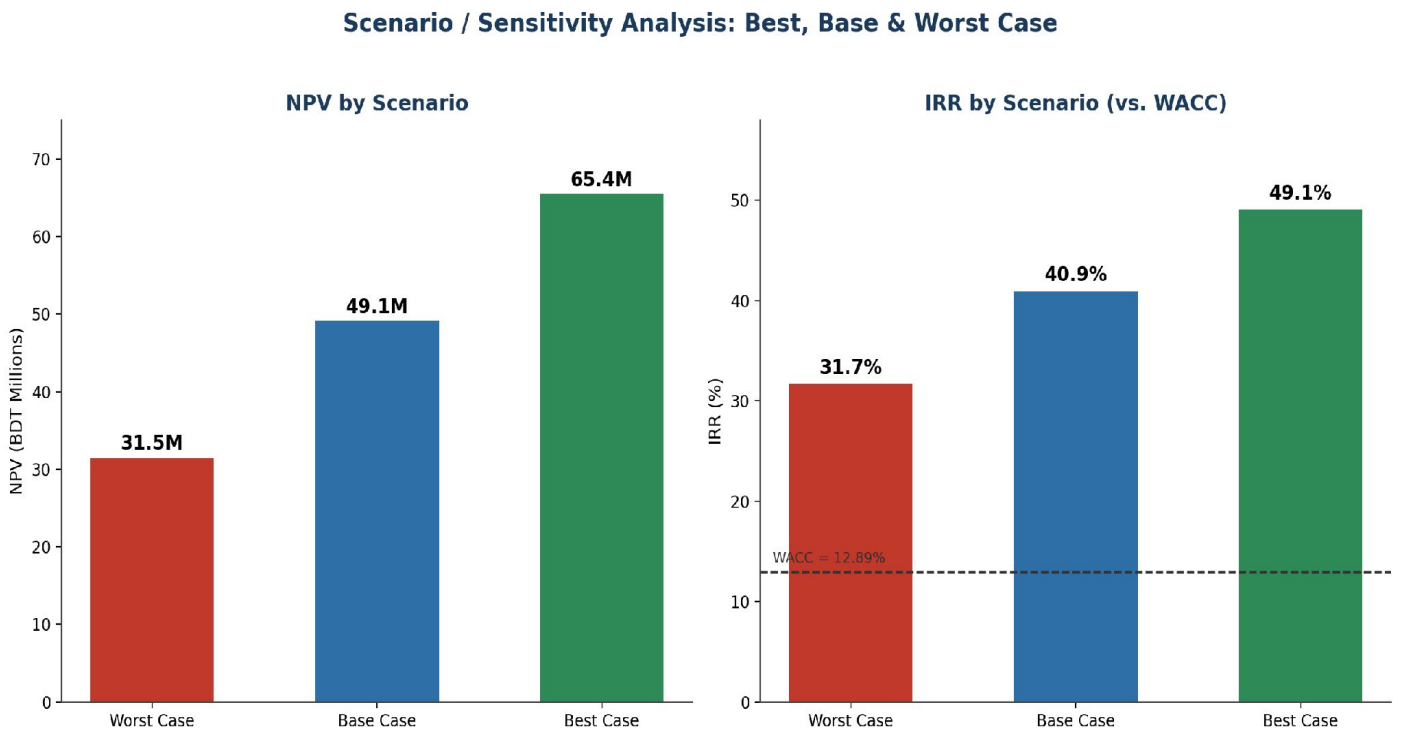
Interpretation

- PI = Present Value of future cash inflows ÷ Initial investment.
- A PI of 2.64 means roughly BDT 2.64 of present value per BDT 1 invested.
- Decision rule: PI > 1.0 → Accept.

Combined with NPV, IRR and payback, PI reinforces the acceptance decision.

Scenario analysis: the decision survives three operating environments

Part 7



Scenario	Revenue change	Opex change	NPV	IRR
Best	+15%	−5%	**65.43M**	**49.07%**
Base	As forecast	As forecast	**49.11M**	**40.93%**
Worst	−15%	+10%	**31.48M**	**31.71%**

All scenarios remain above WACC and positive on NPV.

Provided scenario / sensitivity analysis

Risk view: downside case remains investable, but execution risk is rea

Part 7

Worst-case economics

NPV

BDT 31.48M

Still positive

IRR

31.71%

Still > 12.89% WACC

Modeled adverse drivers

- Revenue −15%
- Opex +10%

Residual risks not fully quantified

- Regulatory changes in parking enforcement
- Construction / installation delays
- Technology and vendor reliability
- Demand adoption and pricing resistance
- Maintenance or wage inflation above plan

Takeaway **The financial thesis is resilient, but final investment should be gated by execution validation.**

Recommendation: ACCEPT

Part 8

ACCEPT

Base case

NPV > 0
IRR > WACC
Payback is short
Worst case still positive

01	Value creation	NPV ≈ BDT 49.1M
02	Return	IRR 40.93% vs 12.89% WACC
03	Recovery	Payback 2.66 years
04	Resilience	Worst-case NPV ≈ BDT 31.5M

Thank You

Questions & Discussion

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